

Spreading 41 Points in the Plane

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Abstract

We place 41 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 38.70712+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 42-point configuration gives 39.71826; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 41 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 38.70712\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 41$ is not on the page. Deleting a point from our 42-point configuration gives 39.71826; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n41.txt
```

in <https://github.com/Ge-limin/math>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
38707126549575678453274241
denominator:
999999999998157502523977
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

41 points, x and y per line. File: friedman-maxmin2d/coords/n41.txt.

```
2.483027092584 -1.078528657570
4.196541351648 1.019121246273
2.497682784428 3.397635523300
1.610627086961 1.726948410597
3.819392074611 1.945273729847
1.457730054610 0.134047550877
0.611420343415 -0.398643609870
2.446544279402 2.275803990310
0.570098494250 3.823952436109
-0.103945964431 0.300106020254
3.270535953607 0.469154551641
2.540629916818 -2.077245108647
-0.922406874840 -0.274456194072
1.575637473365 2.767252192258
-1.049353721709 0.717453327039
-1.137708740165 1.713542374621
-1.175877962146 2.712813664359
-0.269062033280 2.291286831102
-0.271689355667 -1.033776093877
-0.341944716946 3.413089638887
3.376308563844 -1.528026391828
0.615416899895 1.824706702112
1.563824806264 -1.863114830556
2.434618761166 -0.079701028072
-0.230892811299 1.292015541365
1.568488289701 3.767226636518
2.967203299721 1.422039277663
0.659828674833 -1.397471239368
-0.258889777436 -2.179858023155
-1.957798205345 0.125073071382
2.131286107280 0.873183697950
3.327900232426 -0.529198762330
3.298733054292 2.799038442494
0.577247677914 2.823977991850
-1.956169650575 1.138980160295
0.717515095069 -2.395805991303
-1.084740200538 -1.615968779896
4.163817424867 0.019656817383
-1.735457719712 -0.856648880091
0.742363746764 0.832797181001
1.506138386027 -0.864780078621
```